

PRODUCT REQUIREMENT DOCUMENT

Caelaris Linux

Next-Generation High-Performance Arch-Based Linux Distribution with Dual Desktop Sessions, Universal Virtualization, and CachyOS-Grade Gaming Stack

Product Name	Caelaris Linux (x86_64)
Version / Target	v1.0-Rolling (Kernel 6.x)
Distribution Base	Arch Linux Rolling Baseline
Document Author	Caelaris Core Architecture Team
Target ISO Size	3.5 GB – 3.9 GB (Zstandard Level 19)
Primary Desktops	KDE Plasma 6 (Wayland) & GNOME 4x (Wayland)
Date Generated	September 2026
Status	Approved for Release / Production

1. Executive Summary & Product Vision

1.1 Vision Statement

Caellaris Linux is engineered to eliminate the traditional tradeoff between bleeding-edge software access and day-one usability. While Arch Linux offers unparalleled customization and up-to-date packages, its manual setup and configuration barrier prevent mainstream adoption. Existing turnkey distributions often introduce unwanted bloat, unoptimized kernels, or rigid desktop lock-in.

Caellaris solves this by providing an ultra-responsive, modern, and gaming-optimized computing platform. It features out-of-the-box dual graphical sessions (KDE Plasma 6 and GNOME), instant session hot-switching, deep virtualization guest acceleration (VMware, VirtualBox, UTM/QEMU), complete GPU driver stacks (AMD, Intel, NVIDIA), and CachyOS-grade performance tuning without bloat.

1.2 Value Proposition

- **Instant Gratification:** Zero-setup live preview with full 1080p dynamic auto-resizing, pre-configured live user, and full sudo privilege.
- **Freedom of Desktop Choice:** Boot into either KDE Plasma 6 or GNOME from the bootloader, switch seamlessly in 2 seconds via a desktop shortcut, and install both to disk via the Calamares installer.
- **CachyOS-Grade Gaming Performance:** Pre-tuned kernel sysctl parameters (huge map counts, low swappiness, BBR congestion, CAKE packet scheduling), PipeWire low-latency audio, GameMode, MangoHud, Steam, and VKD3D.
- **Universal Hardware & Hypervisor Integration:** Automated hardware driver detection for physical PCs (AMD, Intel, NVIDIA with proprietary utils and DKMS) alongside guest integration suites (VMware `open-vm-tools`, VirtualBox guest services, UTM/SPICE agents).
- **Compact Footprint:** Highly optimized SquashFS packaging with Zstandard level 19 compression, keeping the full-featured dual-desktop ISO strictly between **3.5 GB and 3.9 GB**.

2. Target Personas & Success Metrics

2.1 Target User Personas

User Persona	Key Needs & Pain Points	How Caelaris Addresses It
The Linux Gamer	Needs maximum FPS, smooth frame times, no shader compilation stutter, working Steam/Proton/Wine, and easy performance monitoring.	Pre-configured <code>vm.max_map_count=2147483642</code> , Steam, MangoHud, GameMode, 32-bit Vulkan drivers, and low-latency audio stack out-of-the-box.
The Power User / Dev	Demands latest packages, fast compilation, AUR software access, modern CLI tooling, and responsive terminal environments.	Integrated <code>yay</code> AUR helper, <code>caelaris-install</code> CLI utility, modern Zsh/Bash with auto-completions, Fastfetch, btop, and Git.
Virtualization & Evaluation Tester	Evaluates Linux inside VMware Workstation, VirtualBox, or UTM (macOS Apple Silicon/Intel) and hates manual resolution fixes or clipboard breakage.	Pre-activated <code>open-vm-tools</code> , <code>vboxservice</code> , <code>spice-vdagent</code> , and <code>caelaris-autoresize</code> ensuring immediate 1920x1080 resolution and two-way clipboard sync.

2.2 Key Performance Indicators (KPIs) & Success Criteria

Metric	Target SLA	Implementation Strategy
ISO Image File Size	3.5 GB – 3.9 GB	Zstandard level 19 compression (<code>-Xcompression-level 19</code>), elimination of redundant offline rescue utilities (clonezilla, partclone, darkhttpd).
Cold Boot to Desktop (Live)	< 5 seconds (NVMe SSD)	Elimination of Plymouth graphical locks, pre-created <code>'liveuser'</code> via <code>systemd-sysusers</code> , parallelized asynchronous <code>systemd</code> services.
Display Auto-Resize Latency	< 1 second	Dedicated userspace agent (<code>'caelaris-autoresize'</code>) detecting VMware and VirtualBox monitor geometry changes.
Installer Completion Rate	> 99% Success	Automated Calamares partitioning, non-blocking post-install cleanup module, and virtual CD ejection script (<code>'caelaris-post-install-reboot'</code>).

3. System Architecture & Technical Specifications

3.1 Architecture Overview

Caelaris Linux utilizes a modular build architecture layered on top of the official Arch Linux `mkarchiso` foundation. All core operating system packages, performance configurations, desktop profiles, and installer logic are separated into isolated layers:

```
caelaris-linux/
├── shared/
│   ├── packages.common      # Unified base, gaming, media & hardware stack
│   ├── pacman.conf          # Multi-mirror package repos + Multilib enabled
│   ├── branding/            # Distro identity, os-release, logos
│   └── airootfs/             # Overlay root filesystem
│       ├── etc/sysctl.d/     # CachyOS-grade gaming & network performance
│       ├── etc/sysusers.d/   # Native systemd liveuser provisioning
│       ├── etc/sudoers.d/    # Wheel and liveuser non-interactive sudo
│       └── usr/bin/           # Caelaris utilities (switch-desktop, autoresize, etc.)
├── profiles/
│   └── kde/                  # Primary ISO profile (KDE Plasma 6 + GNOME dual session)
│       ├── packages.x86_64   # Desktop-specific environment packages
│       └── profiledef.sh      # Zstd-19 compression & file permissions
├── installer/
│   └── calamares/            # Customized graphical installer configuration
│       ├── settings.conf     # Linear wizard pipeline
│       └── modules/          # Partitioning, packagechooser, cleanup, reboot
└── scripts/
    ├── build.sh              # Automated build orchestrator
    └── setup_build_env.sh    # Build host dependency bootstrapper
```

3.2 Core Subsystem Specifications

Subsystem	Included Components	Key Capabilities
Base Kernel & System	Linux Kernel 6.x, Linux-Firmware, SOF Audio Firmware, AMD/Intel Microcode, Btrfs-progs, e2fsprogs, exFAT, NTFS-3G.	Hardware auto-detection, universal filesystem support, CPU microcode security patching on boot.
Graphics & Display	Mesa (Gallium), Vulkan-Radeon, Vulkan-Intel, Intel-Media-Driver, NVIDIA-Uutils, lib32-NVIDIA-Uutils, Wayland, XWayland, xf86-video-vmware.	Native Wayland compositing, zero-tear gaming, Vulkan compute, hardware video decode acceleration (VA-API / VDPAU).
Virtualization Support	<code>open-vm-tools</code> , <code>gtkmm3</code> , <code>vboxservice</code> , <code>spice-vdagent</code> , <code>qemu-guest-agent</code> .	Seamless mouse integration, dynamic resolution matching host window, shared folders, host-guest clipboard.
Gaming Ecosystem	Steam, GameMode, MangoHud (64 & 32-bit), VKD3D (DirectX 12 to Vulkan), Gamescope micro-compositor, Fuse2.	Full Proton / Steam Play capability, per-game process scheduling, in-game FPS and hardware telemetry overlay.
Audio Architecture	PipeWire, WirePlumber, PipeWire-Pulse, PipeWire-ALSA, PipeWire-JACK.	Ultra-low latency audio graph, professional JACK routing, seamless Bluetooth audio switching.
Application Management	Pacman, <code>yay</code> (native AUR helper), Flathub App Store backend, <code>caelaris-install</code> CLI.	Access to over 100,000+ Linux packages via official Arch repositories, Flathub sandbox, and Arch User Repository.

4. Detailed Functional Requirements

4.1 Dual Desktop Environments & Live Session Switcher

Requirement FR-1: The distribution must support both KDE Plasma 6 and GNOME sessions within a single unified ISO, both in live preview and post-installation.

- **Bootloader Menu Selection:** Both systemd-boot (UEFI) and Syslinux (BIOS) bootloader menus present distinct options:
 - `Caelaris Linux (KDE Plasma 6 - Default)` : Loads Plasma 6 on Wayland via SDDM.
 - `Caelaris Linux (GNOME Desktop)` : Loads GNOME Shell on Wayland via SDDM.
 - `Caelaris Linux (Copy to RAM)` : Loads the entire system into RAM for portable high-speed usage.
- **1-Click Desktop Switcher (`caelaris-switch-desktop`):** Users can switch between KDE and GNOME during live preview without rebooting. Launching the desktop shortcut reconfigures SDDM autologin and restarts the display manager in 2 seconds.
- **Post-Install Dual Session:** The installed SDDM greeter presents an interactive session dropdown, allowing users to choose between Plasma Wayland and GNOME Wayland at every login.

4.2 Gaming & Performance Optimization Stack

Requirement FR-2: Out-of-the-box kernel and runtime tuning matching CachyOS performance benchmarks for gaming workloads.

- **Sysctl Tuning (`/etc/sysctl.d/99-caelaris-gaming.conf`):**
- `vm.max_map_count = 2147483642` : Essential for memory-mapped games (Star Citizen, Hogwarts Legacy, DayZ) and complex Wine/Proton titles.
- `vm.swappiness = 10` : Prevents aggressive disk swapping, keeping gaming assets in physical RAM.
- `net.core.default_qdisc = cake` : Enables Common Applications Kept Enhanced queue discipline to eliminate bufferbloat during online gaming.
- `net.ipv4.tcp_congestion_control = bbr` : Low-latency Google BBR TCP congestion control for smooth multiplayer network packet streaming.
- **ZRAM Compressed Swap:** Managed by `zram-generator` with Zstandard compression algorithm, dynamically expanding available memory by up to 2x without disk thrashing.

4.3 Calamares Graphical Installer Experience

Requirement FR-3: Foolproof, accessible installation wizard capable of dual-booting, automated EFI partitioning, and seamless handoff to the installed OS.

- **Partitioning:** Supports Automated Guided Partitioning (Btrfs with subvolumes or Ext4) alongside Manual Partitioning for advanced dual-boot setups with Windows.
- **Automated Cleanup:** Removes live environment artifacts (temporary sudoers files, liveuser accounts, autologin configs) from the target disk while preserving user settings.
- **Automatic Media Ejection:** Upon installation completion, `caelaris-post-install-reboot` issues SCSI ejection commands to virtual and physical optical drives (`/dev/sr0`) before initiating system reboot, ensuring the machine boots directly into the newly installed operating system.

4.4 Universal Virtualization Acceleration

Requirement FR-4: Flawless first-boot experience on hypervisors (VMware Workstation, Oracle VirtualBox, UTM / QEMU on Apple Silicon/Intel).

- **VMware Workstation Integration:** Bundles `open-vm-tools` and `xf86-video-vmware` . Includes one-click Windows registry fix file (`fix_vmware_path.reg`) to resolve host engine path discrepancies.
- **Resolution Auto-Detection:** `caelaris-autoresize` actively listens to hypervisor geometry changes and applies 1080p mode via `kscreen-doctor` / `xrandr` .
- **Two-Way Clipboard & Drag-and-Drop:** Pre-enabled userspace agents for VirtualBox (`VBoxClient`) and UTM/QEMU (`spice-vdagent`).

5. Non-Functional Requirements & Quality Standards

5.1 Reliability & Stability

- **No Display Hangs:** Complete removal of Plymouth splash daemon ensures that virtual DRM controllers (`vmwgfx` , `vboxvideo`) are never locked, preventing SDDM initialization freezes.
- **Robust User Provisioning:** `liveuser` is provisioned via native `systemd-sysusers` at early boot, eliminating race conditions where autologin triggers before the user exists.
- **Graceful Network Handling:** Boot scripts contain zero blocking synchronous network queries; the system boots to the desktop in under 5 seconds regardless of network availability.

5.2 Security & System Permissions

Component	Permission / Configuration	Security Rationale
Live User Sudo	<code>liveuser ALL=(ALL:ALL) NOPASSWD: ALL</code> (Mode 0440)	Allows testing hardware tools and partitioners without password barriers in live preview.
Installed System Sudo	<code>%wheel ALL=(ALL:ALL) ALL</code> (Mode 0440)	Standard secure Linux multi-user privilege model; requires user password authentication.
Shadow File Access	<code>/etc/shadow</code> & <code>/etc/gshadow</code> (Mode 0400)	Restricts read/write access strictly to the root user.
Desktop Shortcuts	<code>metadata::trusted true</code> (Mode 0755)	Prevents security untrusted dialogs when launching installer or switchers in KDE/GNOME.

6. Project Roadmap & Milestone Delivery

Milestone	Deliverables	Status & Validation	Target
Phase 1: Foundation	Archiso base, KDE Plasma 6 profile, pacman mirrors, base package definitions.	COMPLETED Tested in QEMU & VirtualBox.	Q3 2026
Phase 2: Dual DE & Installer	Calamares wizard, GNOME 4x integration, 1-click desktop switcher, 30s bootloader timeouts.	COMPLETED Verified live preview & dual install.	Q3 2026
Phase 3: Gaming & Hardware	NVIDIA/AMD/Intel driver stack, VMware tools, sysctl gaming tweaks, yay AUR helper, Zstd-19 size optimization.	COMPLETED Built & validated under 3.9 GB threshold.	Q3 2026
Phase 4: Custom Kernel & Polish	Custom Zen / CachyOS BORE scheduler kernel, bespoke Caelaris dark theme, and automated release mirror network.	ROADMAP In active design and planning.	Q4 2026

Conclusion: Caelaris Linux establishes a modern, uncompromising standard for Arch-based Linux distributions, delivering gaming excellence, hypervisor agility, and desktop versatility in a polished, production-grade ISO.